

DECODELABS CYBERSECURITY INTERNSHIP - PROJECT 4

SYSTEM VULNERABILITY CHECKLIST

Identifying Basic Security Vulnerabilities in a System

by Sanya Zuberi

WHAT IS A SECURITY VULNERABILITY?

A security vulnerability is a weakness in a computer system that attackers can use to gain unauthorized access or cause damage.

Purpose of the Checklist

- Identify common security weaknesses
- Improve system safety
- Reduce cyber risks

WEAK PASSWORD CHECK

Checklist

- ☐ Passwords are at least 8–12 characters
- ☐ Passwords contain letters, numbers, and symbols
- ☐ Default passwords are changed
- ☐ Multi-factor authentication (MFA) is enabled

Risk

Weak passwords can allow hackers to access accounts easily.

SOFTWARE UPDATE CHECK

Checklist

- ☐ Operating system is updated
- ☐ Antivirus software is updated
- ☐ Applications are regularly patched
- ☐ Automatic updates are enabled

Risk

Outdated software may contain known vulnerabilities.

USER PRACTICES CHECK

Checklist

- ☐ Users avoid suspicious emails
- ☐ Users do not click unknown links
- ☐ Files are downloaded from trusted sources
- ☐ Employees receive cybersecurity awareness training

Risk

Unsafe user behavior can lead to phishing attacks and malware infections.

NETWORK SECURITY CHECK

Checklist

- ☐ Firewall is enabled
- ☐ Secure Wi-Fi password is used
- ☐ Unused ports are disabled
- ☐ VPN is used for remote access

Risk

Poor network security can expose systems to unauthorized access.

DATA PROTECTION CHECK

Checklist

- ☐ Important data is backed up regularly
- ☐ Sensitive files are encrypted
- ☐ Access permissions are restricted
- ☐ Backup recovery is tested

Risk

Data loss or theft can damage the organization.

FINAL CHECKLIST SUMMARY

SECURITY AREA	STATUS
PASSWORD SECURITY	☐
SOFTWARE UPDATES	☐
SAFE USER PRACTICES	☐
NETWORK SECURITY	☐
DATA PROTECTION	☐

CONCLUSION

Cybersecurity vulnerabilities can exist in any system, but many risks can be reduced through regular security checks and safe practices. The checklists help identify common weaknesses such as weak passwords, outdated software, unsafe user behavior, poor network security, and inadequate data protection. By following these basic security measures, organizations and individuals can improve system safety, prevent cyberattacks, and protect sensitive information from unauthorized access. Regular monitoring and awareness are essential for maintaining strong cybersecurity.